Automated Promo Discovery for Vertical Microdramas

Problem 1: Find the Promos (Ads) | **Show:** *I Married a Broken Soldier. He was a Duke* (50 Episodes) | **Model:** Gemini 3.5 Flash Lite + Local Air-Gapped Fallback | **Cost:** $0.022 total ($0.0004/ep)

1. Problem Choice, Framing & Core Assumptions

Why Problem 1: Vertical microdramas (1-2 min episodes, 50-100 episodes per series) succeed or fail on user acquisition CTR across TikTok, Instagram Reels, and YouTube Shorts. Manually watching 50 episodes per week creates a critical release bottleneck. Automating promo discovery transforms human editors from manual clip hunters into high-leverage decision makers.

Core Operational Assumptions:

- **Duration & Pacing:** Clips must strictly span **20.0s to 60.0s** with a scroll-stopping hook in the **first 3.0s**. Cutpoints must end on an unresolved emotional peak (cliffhanger) before dramatic catharsis or relief.
- **Confidentiality & The 8-Word Cutoff:** Raw scripts represent proprietary IP. Kuku permits using AI tools on the pack, but public dashboards must prevent IP leakage. *The 8-word dialogue cutoff on our dashboard was our own conservative operational assumption* (not a formal Kuku rule) to display verifiable cutpoints and metrics without leaking full scripts. Full unedited outputs go solely to the private Drive folder.
- **Beat-ID Abstraction (No LLM Math):** LLMs hallucinate floating-point arithmetic. The AI receives numbered beat IDs ([E01-B003]) and outputs beat-ID ranges only. All timestamp math, 20-60s windowing, and hook offset adjustments are handled deterministically in Python.

2. System Architecture & Free-Tier Engineering

We engineered a hybrid deterministic/AI pipeline that enforces invariants in code while leveraging LLMs for narrative appraisal:

Pipeline Stage	Component & Logic	Engine / Runtime	Invariants Enforced
1. Parser	Deterministic regex screenplay parser	Python (0.07s / 50 eps)	Monotonic timestamps, 0 dropped lines, camera cuts preserved
2. Propose	Batched 10 episodes/call candidate proposer	LLM (gemini-3.5-flash-lite)	Outputs beat-ID ranges only; identifies hook & cliffhanger beats
3. Validate	Deterministic boundary & duration repair	Python (<0.01s)	Strict 20-60s window, hook <= 3.0s, whole-beat cuts, dedupe overlaps
4. Judge	Cold-viewer triple gate + 15s post-cut peek	LLM (Cold-viewer prompt)	Hard gates: Hook >= 6.5, Cliffhanger >= 6.5, Standalone >= 6.5
5. Rank	Global listwise ranking over survivors	LLM (1 call)	Cross-series ranking, conversion rationales, spoiler/funnel notes

Solving the Free-Tier Quota Ceiling (<= 12 Calls Total): The Gemini free tier enforces a strict daily limit (~20 req/day). A naive 1-call-per-episode architecture exhausts quota in 5 episodes. We restructured the entire 50-episode run into **5 Propose calls (10 eps/batch) + 4 Judge calls (~30 clips/batch) + 1 Rank call = 10 calls total**. A SHA-256 disk cache (.cache/llm_cache) and per-batch checkpoints ensure crashes resume without burning quota. Re-running costs 0 calls and runs in 0.06s.

3. Provider Data Privacy & Terms Audit

Creative screenplays are valuable trade secrets. We audited provider data retention terms prior to running:

Provider	Data Logging & Training Terms	Production Assessment
Google Gemini (Free)	Prompts and outputs may be logged and reviewed by human reviewers to train models.	Fine for rapid technical prototyping; NOT acceptable for confidential em
OpenAI API	Business terms: API inputs/outputs are NOT used for model training (opt-out by default).	Safe for commercial screenplay extraction under standard commercial
Anthropic Claude API	Commercial terms state customer API data is never used to train generative models.	Safe for production screenplay workflows.
OpenAI-Compatible (Local)	100% on-premises / air-gapped (Ollama / vLLM). Zero external data transfer.	Optimal for confidential pre-release IP. Fully supported in our app & en

4. Quality Evaluation: Human Ground Truth Benchmark & Real Failures

We hand-curated gold-standard editor promo cuts across Episodes 1-10 in eval/ground_truth_user.json based on professional vertical ad pacing. Evaluation strictly enforces a **>= 50% temporal IoU threshold** against the gold clips:

Benchmark Metric (n = 10 Episodes)	Heuristic Baseline (Regex)	Production AI Pipeline (Gemini)	Editorial Takeaway
Top-3 Hit Rate (>= 50% IoU)	80.0% (8 / 10 hits)	90.0% (9 / 10 hits)	Pipeline discovers nuanced character drama missed by keywords.
Mean IoU on Hits	0.730	0.896	AI cutpoints align tightly with professional editor boundaries.
Exact 100% Matches (IoU = 1.000)	0 / 10	2 / 10 (E01 & E09)	AI perfectly captured cathedral betrayal and sick child emergency.
Run-to-Run Stability (Repeatability)	100.0%	100.0%	Beat-ID boundary anchoring guarantees zero cutpoint jitter.
Baseline vs Pipeline Agreement	—	33.8%	AI explores non-obvious emotional beats beyond loud sound effects.

Per-Episode Evaluation Breakdown (Strict >= 50% IoU):

Episode	Gold Standard Scene Summary	Baseline Result	Pipeline Result	Pipeline IoU	Status
E01	Cathedral altar betrayal ('Turn around, Juliana')	Hit (IoU 0.676)	Hit (00:11.3 - 00:43.1)	1.000	Exact Match
E02	Gerald's brutal rejection ('Far too plain...')	Hit (IoU 0.677)	Hit (00:00.0 - 00:44.7)	0.906	Excellent
E03	Scarred warrior knee-drop marriage proposal	Miss (IoU 0.000)	Miss (01:09.9 - 02:09.9)	0.264	Known Failure
E04	Gerald mocking outcast bride & defiant exit	Miss (IoU 0.271)	Hit (00:01.5 - 00:58.8)	0.815	Outperformed
E05	Stepmother hurls mother's ledger into mud	Hit (IoU 0.713)	Hit (00:04.2 - 00:58.2)	0.931	Excellent
E06	Rainy woodland cabin arrival & exile	Hit (IoU 0.719)	Hit (00:00.0 - 00:40.0)	0.963	Near Exact
E07	Aristocrat carriage confrontation in mud	Hit (IoU 0.920)	Hit (00:19.5 - 01:17.5)	0.836	Excellent
E08	Mother's botanical grimoire & battle scars	Hit (IoU 0.519)	Hit (00:03.4 - 00:51.2)	0.754	Strong
E09	Frantic night pounding for dying child	Hit (IoU 0.612)	Hit (00:00.0 - 00:48.2)	1.000	Exact Match
E10	Warrior returns bloodied; intimate care	Hit (IoU 0.805)	Hit (00:00.0 - 00:57.6)	0.833	Excellent

Where the AI Got It Wrong (Honest Failure Analysis):

- **1. The Episode 3 Boundary Drift (The 0.264 Miss):** The editor selected the stranger's knee-drop marriage proposal (01:09.9 - 01:45.8, 35.9s). The AI proposed a 60.0s window continuing into crowd chatter. While it caught the hook, excess tail padding dragged IoU down to 0.264. *Fix: add an action-decay detector to cut immediately upon proposal delivery.*
- **2. Lore-Dependent Dramatic Irony (Episode 8):** In Episode 8, the AI proposed a quiet dialogue about herbs because it foreshadowed that the broken soldier was secretly the Duke. To a cold TikTok scroller, this was boring gardening. *Fix: cold-viewer judge gate strictly penalizes backstory-dependent tension.*
- **3. Mid-Sentence Cut Truncation:** In early unconstrained tests, models cut dialogue mid-word (*'Take your...'*) to satisfy duration limits. *Fix: code strictly snaps cutpoints to whole beat boundaries (beat.end).*
- **4. Batched Judge Anchoring Bias:** Evaluating 30 candidate clips in a single prompt caused relative grading (grading clips against each other rather than absolute viral ad potential). *Fix: explicit prompt fencing per candidate.*
- **5. Score Clustering Audit:** LLM judges frequently clump scores at 8.0. In our real run, **only 2.0% (2/99) scored exactly 8.0** (mean 6.77, pass rate 55.6%), proving robust calibration without grade inflation.

5. Scalability to Full Production & Top Discoveries

Global Top Promo Across 50 Episodes (#1 Ranked — Episode 45, 00:54.00 - 01:20.40 | 26.4s):
• **Hook Beat (E45-B024):** *"You poisoned my mother. You tortured my wife."* (Hook rating: 8.5/10)
• **Cliffhanger Beat (E45-B032):** *"Lady Juliana Ashcroft is my lawful wife... and the Duchess of Ashburton!"* (Cliffhanger rating: 9.0/10)
• **Composite Score: 8.8 / 10.0** | **Acquisition Impact:** Immediate high-status confrontation with instant villain unmasking.

Scale-Up Economics: Processing Kuku's weekly slate of 10 new 50-episode shows costs **$0.22/week ($11.44/year)** on cloud flash models, or **$0.00** via on-premises vLLM clusters. The prototype in app.py provides a full public dashboard and a self-service runner supporting OpenAI-compatible endpoints.

Submission Artifacts Verified: *Full unedited production outputs (50 JSONs, run_meta.json, all_episodes_promos.json) in outputs/raw/. All mock runs strictly excluded. Unit tests (tests/test_parser.py, test_pipeline.py): 6/6 passing.*